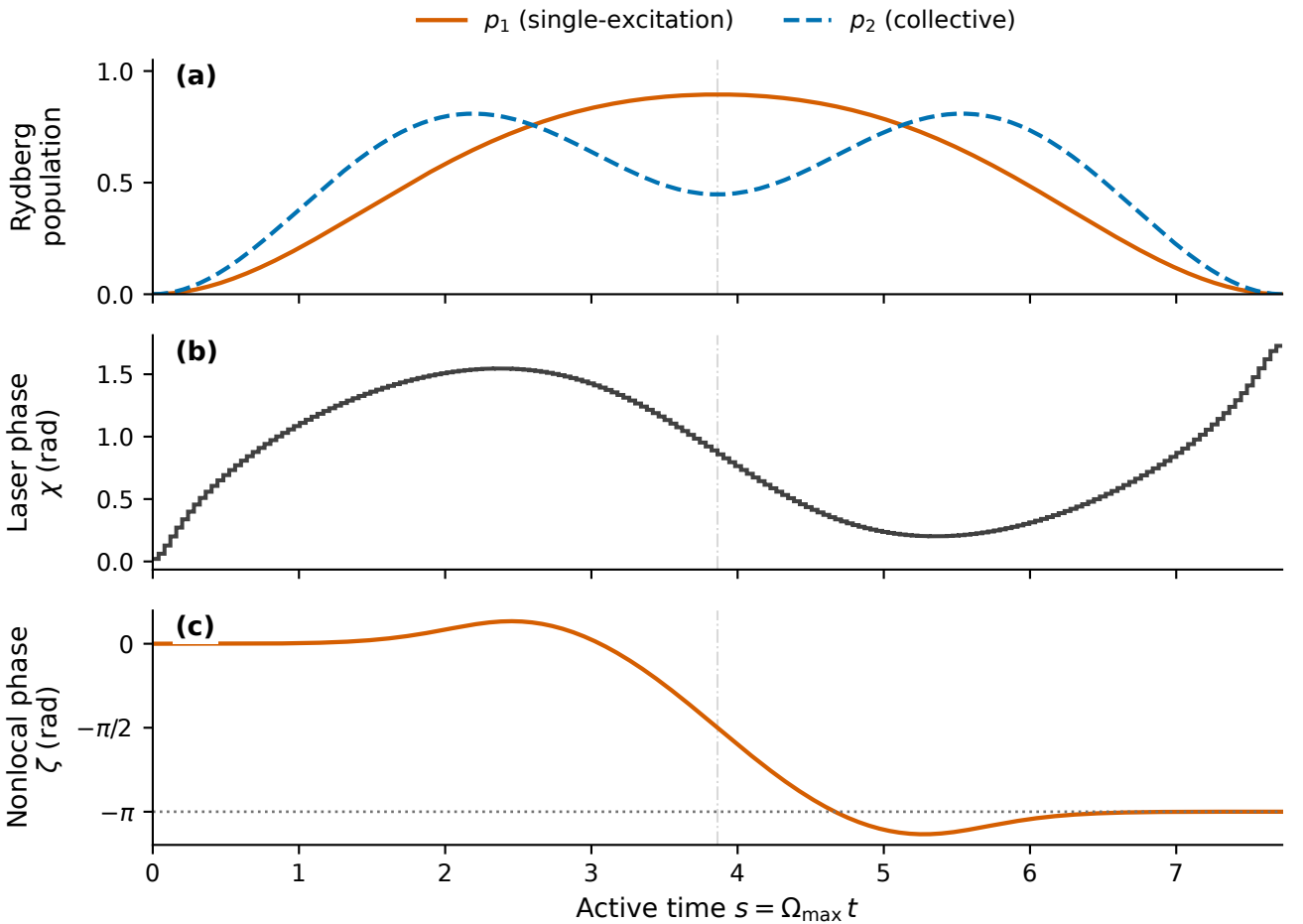
p_1 (single-excitation)
p_2 (collective)
(a)
Rydberg population
1.0
0.5
0.0
(b)
Laser phase χ (rad)
1.5
1.0
0.5
0.0
(c)
Nonlocal phase ζ (rad)
0
$-\pi/2$
$-\pi$
0
1
2
3
4
5
6
7
Active time $s = \Omega_{\max} t$